“Niñas Atómicas” (Atomic Girls) Hands-On Detector Assembly Instructions

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1 Hands-on Instructions

- Wear gloves to first wrap the plastic scintillators in aluminum foil.
- Place the scintillators on top of the SiPMs. When doing this, the uncovered side should be touching the sensor, and the aluminum covered side should point to the centers of the main case when placing the SiPM inside the hexagonal casings. A small amount of lubricant is applied to the central square of each SiPM when placed inside the hexagonal casings.
- Connect the SiPMs to the PCB, where two 3-connector female-to-female cables are needed. Two pins correspond to the power supply voltage and the other one corresponds to the measured signal. When connecting the cables that link the scintillators to the rest of the circuit, the opening of the detector cases should be aligned with the SiPMs cable entries. Screws can be used to secure the cover of the casings after placing the cables. It is important to keep track of which cable corresponds to ground “GND”, signal “S” and positive “+” on the SiPMs before closing the detector, as these need to match the “GND”, “Signal” and “H_Voltage” entries on the PCB as labeled in Figure 1. The pins “GND” and “S” are labeled on the right and left of the SiPMs.

On the PCB, the ground is the unlabeled entry right below the labels "SiPM 1" and "SiPM 2".

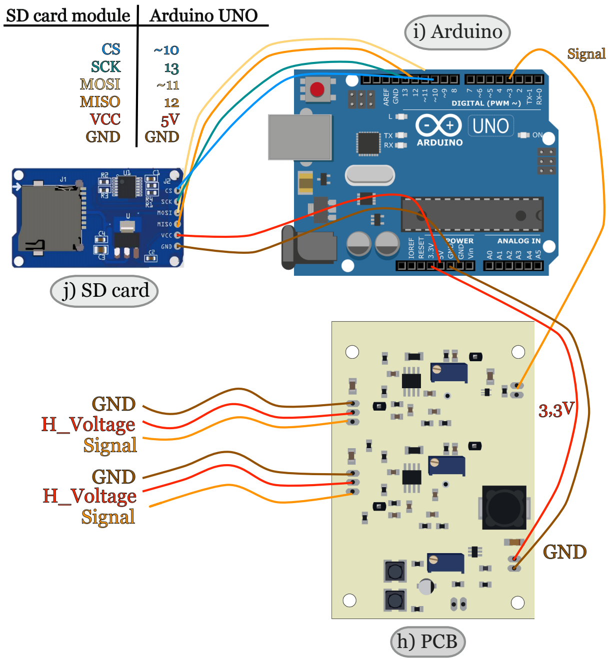

Figure 1: Diagram showing the electrical connections of our muon counter detector experiment [Cottin and Garay(2025)].

- Connect the PCB to the Arduino, for which three female-to-male cables are needed. Two of them correspond to the power supply voltage and the other one corresponds to the measured signal (with "GND", "3,3V" and "Signal" labels in Figure 1). The signal, after being processed by the PCB, comes out through the port labeled "TTL out". Using a female-to-male cable, this output is connected to the digital input D3 labeled "∼3" on the Arduino. This is to supply power to the circuit.

- Connect the Arduino to a computer with a USB cable (or directly to a power socket with a transformer if needed).

- Prepare the multimeter and calibrate the voltage of the potentiometers of the PCB, which correspond to the three blue boxes on the PCB. When measuring voltages with the multimeter, one of the probes is always touching a ground point on the PCB ("GND" label in the lower right corner of the PBC) and the other to the point on the PCB where the voltage is to be measured. These measurement points correspond to the circles labeled "TCom1" and "TCom2" right below potentiometers (1) and (2), and towards the left of the "29.4 V" label below (3). For potentiometers (1) and (2) the voltage increases when turning the screw clockwise, and should be set to 2.18 V. For potentiometer (3) the voltage increases when turning the screw counterclockwise and should be set to 29.4 V.

- Six more cables are needed to connect the SD card reader to the Arduino, with "GND-GND", "VCC-5V", "MISO-12", "MOSI-∼11", "SCK-13" and "CS-∼10" connections, detailed in Figure 1.

  A "step-by-step" tutorial in Spanish created by our tutors Mariel Poduje and Devika Mukhi can be found in [Niñas Atómicas(2025)].

- After assembly, install the Arduino IDE Software in their computers, following the instructions in Ref [Arduino(2025)]. The program registers an index for each muon together with the time (in seconds) each muon was detected. A `.txt` file will be saved to the SD card with this data.

References

[Cottin and Garay(2025)] G. Cottin and F. Garay, (2025), arXiv:2511.05754 [physics.ed-ph] .

[Niñas Atómicas(2025)] Niñas Atómicas, "Niñas atómicas github repository," https://github.com/gfcottin/Atomicas (2025).

[Arduino(2025)] Arduino, "Arduino ide," https://www.arduino.cc/en/software/ (2025).